WebTV™ Service Installation Guide

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Preface

This book describes how to install and configure the hardware and software for the WebTV™ service. This current preliminary draft of the *WebTV Service Installation Guide* is provided for reference only.

Caution: The information contained herein is preliminary and has not been tested or verified.

This Guide Contains

Chapter 1, "Introduction," provides a brief overview of the WebTV system and operation.

Chapter 2, "Ordering Services and Equipment," describes the facilities, hardware and software that must be purchased to install and configure a WebTV service.

Chapter 3, "Hardware Installation," describes how to install the hardware that makes up a WebTV service.

Chapter 4, "Configuring the Systems and Software", describes how to configure the software to enable a WebTV service.

Conventions Used in This Guide

The following typographical conventions are used in this book:

Typeface	Description	Example
`AaBbCcDd`	The names of tags, attributes, directories, on-screen computer output	The WebTV interface implements the `<!-->` comment tag.
AaBbCcDd	Placeholder: replace with a real name or value	`<a href="`*attribute*`">`
AaBbCcDd	Book titles, new words or terms, or words to be emphasized	Using the *WebTV Internet terminal*, consumers...

Chapter 1

Introduction

This chapter provides a brief overview of the WebTV system and operation. The following topics are covered:

- The User's Experience with a WebTV Internet terminal
- How the Service Software Works
- The WebTV Service Architecture

Note: A more complete description of the WebTV system and software is described in the "*WebTV System Software Overview*."

The User's Experience with a WebTV Internet terminal

A WebTV customer buys the Internet terminal having been promised two things: it will provide Web access and electronic mail service. The Web access can be configured to prohibit connection to sites appropriate only for mature browsers.

WebTV strives to minimize demands on the customer; the whole experience should require thinking about only three things: the Internet terminal, the 'Net, and the bill. In particular, the ideal experience is one where the customer never thinks about the WebTV service infrastructure or the support behind it.

The customer unpacks the Internet terminal, plugs it into an outlet, plugs it into a TV, and plugs it into a phone jack. He or she turns it on, waits a short time, and responds to a few questions. These request a a check or credit card number for paying a monthly Internet access fee, and give the customer an opportunity to create separate accounts. Each account can have a password to ensure privacy when sending and receiving email, and each can be "content-restricted"--denying access to Web sites deemed inappropriate for certain users.

A customer can use WebTV to access the entire World Wide Web, but can also choose to view content made available for WebTV users only. This material is updated frequently to suggest interesting sites and to provide an easier interface than typing URLs. When done with a session, the customer merely turns off the Internet terminal. To begin surfing again, she turns it back on.

The customer can also move the Internet terminal from its initial location anywhere in the United States to any other location in the U.S. Each time the Internet terminal is reconnected to power, it asks the customer whether its phone number has changed. If so, the Internet terminal takes a little more time to start up, but the customer does not have to enter any other information.

How the Service Software Works

This section briefly describes how the service software works.

Connecting the Internet terminal to the Internet

When the client Internet terminal is first connected, it dials a toll-free number which is answered by a bank of modems in Palo Alto, California. These modems establish a 28.8 Kilobit per second PPP connection with the client Internet terminal, which then establishes a TCP/IP connection to a `preregistrationd` process running on a UNIX machine in Palo Alto.

The `preregistrationd` process verifies that the call is coming from a WebTV client Internet terminal and then connects to an ANI-server. The ANI-server uses Automatic Number Identification to determine where the Internet terminal is calling from and returns that information to `preregistrationd`.

`preregistrationd` uses that data to download information to the client Internet terminal. It sends a small program written in TellyScript, which contains the appropriate (cheapest) local phone numbers of an Internet provider. It also sends an encryption key to be used when communicating with WebTV. Finally, it sends the IP addresses and TCP port numbers of the other processes in Palo Alto that the client will need to talk to. When the download is complete, `preregistrationd` sends the encryption key and ANI information to the Customer Database.

With pre-registration complete, the Internet terminal disconnects from WebTV's toll-free number, and uses the TellyScript program to make a local call for an Internet connection to WebTV. It tries two phone numbers twice each; if these go unanswered or are busy, it calls another toll-free WebTV phone number and makes its connection in Palo Alto.

Whenever the Internet terminal loses and then regains electrical power, it asks the customer whether it is connected to a different phone number. If the customer answers “Yes”, the Internet terminal repeats the pre-registration process.

When WebTV wants to provide a new or different local phone number for an Internet access provider, a new TellyScript program can be downloaded. (Phone numbers can be changed to balance loads between access provider locations, or to incorporate new providers or locations.)

Receiving the Connection at WebTV

Whenever the WebTV Internet terminal connects to WebTV's service, it asks first (through TellyScript) for a `headwaiterd` process. All WebTV processes come in pairs, so if for some reason the client Internet terminal's chosen `headwaiterd` process doesn't respond, the client can ask for its second choice.

On a Internet terminal's first Internet connection, `headwaiterd` connects it to a `registrationd` process. This process asks the customer to create a user name and password, and requests billing information, including a credit card number. The credit card number and balance are checked through a commercial service. If the customer's credit limit can pay the monthly fee, then `headwaiterd` stores the information in the Customer Database and creates a directory on a file server to store the user's email and Web-specific information.

On subsequent connections, `headwaiterd` and the Radius server combine to authenticate the Internet terminal using its serial number and its “shared secret”. `headwaiterd` then checks the status of the customer's account, asking if necessary for new billing information. If the client needs an update, `headwaiterd` asks a process called `flashromd` to provide it. Finally, `headwaiterd` passes a new set of process addresses and configuration values to the client Internet terminal. It distributes the addresses in round-robin fashion, providing a simple but effective load-balancing facility.

WebTV Services

To the customer, WebTV provides two major services: Web access and electronic mail. (Network news and chat services are planned for future releases.) The principal process behind Web access is `proxyd`, which handles http requests from the user, issues them through one of WebTV's Internet Access Providers, and returns the results. In addition, it caches frequently accessed Web pages for faster response, compacts certain kinds of data, refuses to transfer adult data when the client account is content-restricted, and translates certain pages which would otherwise report errors to the customer.

When WebTV needs to conserve bandwidth or processing power, the client Internet terminal can be told to contact Web sites directly, rather than going through WebTV and `proxyd`. This stopgap solution can't be used when the client is content-restricted, however, so the `proxyd` processes must still handle those requests.

Similarly, if the client can't reach WebTV, but isn't in KidSafe mode, it can also contact sites directly. It loses, however, the performance advantages of caching and compacting provided by `proxyd`.

Web access is facilitated by two other processes, called `cookied` and `favorited`; cookied stores and retrieves user preference data--called “cookies”--that certain sites expect the client to retain, and favorited maintains the customer's list of favorite sites--often called “bookmarks” by other browsers. Both of these clients depend on a file server, called “NetApp” because it's implemented on a Network Appliance server, for fast storage and retrieval of this data.

The WebTV Service Architecture

This section describes the servers and networks the comprise the WebTV service architecture.

Servers

The processes that implement the WebTV service run on UNIX servers which are configured and networked for optimal performance and prudent security.

The `proxyd` processes run on proxy servers, the `mailtod` processes run on mailtod servers, the `notifyd` processes run on notify servers, the aliasd processes run on postoffice and mailsorter servers, and almost all the other service processes run on the storefull and storeless machines. The latter two categories differ in their need for connectivity to shared file services. For example, the storefull servers need file service because they run `cookied` and `favorited`, which store and retrieve user-specific pieces of data.

A source file called `arcadia.cf` maps each server to its role, and each role to the set of processes it must execute. An admin server runs a `monitord` process which ensures that the appropriate processes on each server are restarted if necessary, and special servers are dedicated to ANI service, harvesting and processing log files, and running the customer and service databases.

Networks

The WebTV service is made up of several networks, each of which connects only those machines needed to provide a related set of services.

The iapisp net

The heart of the WebTV service is the iapisp network. (The name stands for Internet Access Provider/Internet Service Provider, a distinction whose importance to Operations is fading.) This network forms the link between the Internet router(s) and every server which might have to communicate with the client. Thus, all proxy, storefull, storeless, mailtod, mailsorter, postoffice, and notify machines are connected to this network.

If some of the other processes run on dedicated machines, those machines also have an interface to the iapisp net. Examples have included ns servers for DNS service, realaudio servers to run realaudio, and prereg servers for `preregistrationd`. In other cases, these processes have been assigned to share the storefull, storeless, and other servers.

Finally, the monitoring and administration machines, the admin and log processing servers, are connected to iapisp.

The store net

Many of the WebTV service servers make no use of access to shared or permanent storage. Each proxy server, for example, has a large amount of disk storage attached to hold its cached data. Some servers, however, need to store and retrieve electronic mail, cookies, and favorites. Because this data needs to be accessible to all of postoffice, `mailtod`, and storefull servers, and because it needs to persist between customer sessions, it is stored on dedicated file servers.

The file servers are Network Appliance machines, which are optimized to provide extremely fast NFS service. A WebTV service dedicates one NetApp to mail storage, and one to storing other data. The “store net” connects these two file servers with the production servers that need file service.

Keeping the store net separate from the iapisp net has two major advantages: high-speed NFS access does not compete for bandwidth with the rest of the service transactions, and private user data is one further hop away from inappropriate attempts to compromise it.

The router-nexus net

There are other networks needed to run the WebTV service; a single network called the “router-nexus” serves as the central connection point where these networks meet the iapisp network. In general, only routers are connected to the router-nexus; the Internet router(s) are connected here, but no other iapisp machines have interfaces on the router-nexus.

The db net

There are two hosts assigned to running the database applications that support the WebTV service: custmaster and custservice. These machines are connected to each other by the “db net”. This network is attached to the router-nexus by two small firewall routers. These routers' access lists provide an extra level of security for the database machines.

The db-admin net

A third machine is dedicated to providing backup and redundancy services to the two database machines. The three are connected on a high-speed network called the “db-admin” net.

The admin net and console service

Access for administration of the machines on the iapisp and store nets is provided in two ways. Each machine has an interface on the “admin net”, which also hosts several systems dedicated to administration and monitoring of the service machines. The admin net is also connected to a router which serves as a gateway to the internal network of the company.

In addition, each production machine and each of the routers, network switches and hubs are connected through its console port to a remote terminal server.

One idea under consideration is to use the capabilities of the switches which implement the admin net to provide an extra level of security. In this model, the administrative hosts would be able to reach all machines on this net, but the iapisp and store machines would only be able to communicate to hosts in their own classes. If this approach is taken, the admin net could be extended to serve the database machines.

The modem net

US Robotics modem banks are used to hold the modems which provide both the pre-registration service and the backup access for customers whose local Internet providers couldn't be contacted. The network which hosts the modem banks is connected to router-nexus.

Software

This section is still being developed, and will include information on the following topics:

- WebTV
- Introduction to “service groups”
- Radius
- sendmail

Oracle and Applications

Databases are used to store billing information (credit card numbers), distillations of phone records, URLs visited (someday)—probably only aggregated, name-to-subscriber-id mapping, logins and authentication— up to 6 usernames per device—which have to be authenticated.

Support Servers

This section is still being developed, and will include information on the following topics:

- Admin
- Harvester

  (We harvest everything, and store it on a harvester NetApp and other disks and devices.)
- Database Servers

Security

This section is still being developed, and will include information on security concerns, including:

- access to our resources
- abuse of our resources
- privacy protections for users (while protecting the value of information we could resell)

Chapter 2

Ordering Services and Equipment

In a schedule for building a WebTV service, the most time can be spent simply in waiting for the equipment, software, and services that make up the system. This chapter specifies what's used in building the service, and captures some of the reasoning behind those specifications.

This chapter covers the following topics:

- Facilities
- Compute Server and Local Disk Menu
- The Machines Used for Each Class of Compute Server
- File Servers
- Tape Drives
- The Number of Machines Needed for Each Machine Class
- Internet Providers
- Internet Router(s)
- Network Device Menu
- SCSI Controller Cards and SCSI Connectors
- Console Servers
- Cables and Connectors
- Modems
- Telecommunications Access
- Software

Facilities

At the most basic level, the computing and network equipment needs a place to sit, where it can receive electrical power, access to telecommunications providers, and appropriate environmental conditions. Little of this information is truly specific to WebTV, except for the emphasis on redundant facilities to ensure the high availability and reliability of the service.

Space

There are as yet few examples of service buildouts; two have been constructed in Palo Alto and are currently in use as production services. These two facilities are referred to as Alma Street and Bryant Street, after the locations of the buildings which house them. Prototype and test versions have also been constructed, including systems designed to fit in a single rack and support most--though not all--of the WebTV service functions.

It is WebTV's intention to develop rules that map the number of users to be supported to the number of computers, modems, racks, cables, and so on, needed to serve those users. Until those rules are observed or derived, however, it is probably sufficient to note that no WebTV service has yet expanded beyond ten racks. The philosophy is that if a service approaches that size, the company is better served by building out a new service in a different location, reducing its dependence on machines, power, and networks shared in a single service.

Racks

In its Bryant Street facility, WebTV chose equipment racks from Bud, outfitted with an 11-outlet power strip, the largest that could be provided. That outlet supports five machines, five disk racks, and a hub or switch. There are both deep and shallow models available, and both were used.

These nice ratios haven't really been used in practice: the proxy machines are using more disk than expected, and Ultra-2s draw more power. As a result, some racks are drawing 13 amps of power rather than the 6 or 7 amps supplied. Those racks are now being fed power from other racks. In future buildouts, each rack should ideally be supplied with power more than sufficient to meet the needs of the equipment it houses.

Patch Panels and Wiring Runs

In the Bryant Street facility, network cabling is run overhead, and connected to a 24-port RJ-45 Category-5 patch panel at the top of each rack. Wiring runs need to be ordered to support cables coming from the telecommunications access facilities, and to support cables running between racks.The cabling connects the racks to two centralized racks which contain patch panels, network devices, and serial console servers.

With this arrangement, a host is added to a rack, and then connected to power, console service, and the relevant networks by plugging into pre-wired outlets and jacks on the rack itself. All that remains is to patch the host's corresponding network jacks on the central panels into the correct pieces of network equipment.

Power

The bulk of the equipment used to run the WebTV service in Palo Alto uses 120-Volt, 60-cycle electric power. The 11-outlet power strips employed have an L5-20 connector on them.

The service must be kept running almost continuously, so continuous electrical power is a necessity. UPS functionality is needed, with an on-site generator behind it. WebTV recommends redundant UPS's, connected in series.

In the Bryant facility there is no raised floor, so both power and network cables are run overhead. The power overhead is 3-phase, with each L-520 on a separate bus bar. This arrangement allows easy examination of the amount of power being drawn by each leg.

A prototypical WebTV service would have five to eight racks served by three 50-amp legs.

Cooling

Cooling facilities must be provided to maintain the functionality of the servers and network equipment. The two 15-ton Lieberts in the Bryant facility are more than sufficient for the 30-40 machines currently deployed there.

A raised floor would perhaps improve cooling efficiency, but a small facility may not warrant the extra expense. Even with a raised floor, WebTV recommends that the network wiring remain above the racks for easier trouble-shooting and maintenance.

Telecommunications Infrastructure

The production facility should, if possible, have multiple connections to the local telecommunications infrastructure. It should be possible to run the equivalent of at least two DS-3 connections to the site, over paths as physically distinct as can be arranged.

High speed terminations of this kind often require space for racks and special provisions for power. In the Bryant facility, for instance, WebTV has arranged for the installation of 50 amps of 48-Volt power with redundant rectifiers. This custom work drives the network cabinets from two local telecommunications providers: MFS and Pac Bell. These companies would have done this work themselves, but then each would have required its own power rack as well.

Compute Server and Local Disk Menu

Machines from Sun Microsystems are used for all servers which run the WebTV service processes. The general approach to service implementation has been to dedicate a server to each compute-intensive process, or to several less demanding processes. This approach allows selection of equipment at very reasonable price/performance ratios, and allows WebTV to buy from a leading vendor whose machines are widely understood by both developers and support staff.

The two models deployed are Sun's Ultra Enterprise-2 Model 2200s ("Ultra-2s"), which have two 200MHz processors, and Ultra Enterprise-1 Model 170s ("170s"), with a single 170MHz processor. The Ultra-2s are purchased with an initial configuration of 256 MB of RAM, in 64 MB SIMs, and supplemented with an additional 384 MB. The 170s are purchased with 192 MB of RAM. Both of these

machines provide excellent performance at a reasonable price; the 170 is the fastest machine available in a single-processor configuration, and has proven sufficient to run most of the WebTV processes under production loads.

All of the compute servers have two 2GB internal disks. In the future, when they become available, a pair of 4GB internal disks would be better: user event logs, service logs, and UNIX system-process accounting logs take up significant space, as do the service processes themselves. The Solaris 2.5.1 that is run on these machines takes up 78 MB at this time, and swap space also increases disk demand.

Dedicated external disk storage is needed for some of the different server classes. Currently, WebTV uses SunSwift cards as disk controllers. These provide a 10/100 auto-sensing Ethernet interface and a Fast Wide SCSI connection to an external disk rack. The disk racks currently deployed are from Adjile and can support up to nine 2.1-GB Seagate ST32155W disks on four SCSI busses. WebTV servers access these racks through three SCSI interfaces, controlling three these disks each. Additionally, these racks require three 300-watt power supplies, because a single power supply is insufficient to spin up nine disks, and redundancy is recommended to guard against failure of a power supply.

Several members of the Operations team believe that a SparcStorage Array might be a better solution. With the current discount from Sun, and the possibility that up to 30GB might be in use on some machines, it no longer seems unreasonable to select a technology which supports two FibreChannel busses and up to 60 GB of disk. In addition, at least one team member reports that three or four out of every ten of the Adjile racks will require repairs when they are delivered.

The Machines Used for Each Class of Compute Server

Proxy Servers

The proxy servers run proxyd, which puts heavy demand on both CPU and disk storage. Each WebTV client connection asks its proxy server process to fulfill its requests for Web pages. The process does GIF-to-JPEG translation, called transcoding, in some cases, which is among its most CPU-intensive functions. It also attempts to cache almost all of the pages it provides; these pages are stored on its local disks. In effect, each proxy server is trying to cache the whole Internet by itself.

Ultra-170s were tried for these servers and proved to be insufficient. Ultra-2s are now used, both for their two 200 MHz processors, and for their additional Sbus slot. This fourth S-bus slot allows WebTV to add a third SunSwift SCSI controller; the proxy servers use all nine disks in an Adjile rack, split across the three SCSI-bus controllers.

Another advantage of the Ultra-2s is that they support up to one gigabyte of RAM. Initial RAM configurations of the proxy servers proved to be suboptimal as the service grew; neither 192 MB nor 320 MB of RAM could keep the machines from paging so much that TCP connections would time out. Adding 384 MB to the 256 MB default configuration of the Ultra-2s seems to have worked, so an Ultra-2 with 640 MB of RAM is ordered for each proxy server, with 3 SunSwift SBus controller cards and an Adjile disk rack with nine disk drives. Each server also needs two 10Mb-Ethernet cards in order to be connected to the iapisp and admin networks.

Mail Processing Machines

The only other machines with external disk are the ones used to handle mail: the mailsorters, postoffices, and `mailtods`. For all of these machines, WebTV uses Ultra-170s with 192 MB of RAM.

The mailsorter hosts are usually purchased in pairs, accompanied by three SunSwift controller cards and a disk rack with six 2-GB disks to hold email received from or destined to the Internet. They need two 10-Mb Ethernet cards to be connected to the iapisp and admin networks.

Both the mailtod and postoffice also have external disk, in case they need to queue mail which temporarily can't be processed or delivered.

postoffices are also purchased in pairs, each using two disks on a four-disk Adjile rack They each need a SunSwift controller card, two 10Mb-Ethernet cards, and a 100-Mb FastEthernet card, because they are connected to the iapisp, admin, and store nets. The disk rack should be purchased with dual power supplies.

Similarly, mailtods are purchased in pairs, again sharing an 8-GB rack. They each also need a controller card, two 10Mb-Ethernet cards and a 100-Mb FastEthernet card, because they are connected to the iapisp, admin, and store nets.

The Storefull Machines

The storefull servers run `cookied` and `favorited`, two processes which depend heavily on NFS access to store and retrieve customer-specific data. Ultra-170s have generally proven sufficient for this purpose, although they have occasionally triggered warnings set to go off when a machine's load average exceeds 2.0.

The storefull machines, like all of the 170’s, are configured with 192 MB RAM. They need two 10Mb-Ethernet cards and a 100-Mb FastEthernet card, in order to connect to the iapisp, admin, and store networks.

Because their access is primarily to data shared with other storefull machines, they are not attached to external disk racks.

Other Services Machines

All of the other service machines--the storeless machines that provide home directory service and WebTV-specific content, the preregistration machines that help a new or moved client re-establish connections and Internet service, the name servers that process DNS names of Web sites and email addresses, realaudio machines that run a realaudio server, news processing machines, and the notify servers that turn on a small light on the client when email has arrived—are basic Ultra-170s. They are upgraded to hold 192 MB of RAM, but have no external disk, relying on the two 2-GB internal disks that are standard. Each needs either two 10-Mb Ethernet cards to be connected to the iapisp and admin networks, and perhaps a third to be connected to the store network.

The Admin Machines

There are three machines dedicated to monitoring and administering the entire set of services machines: admin, harvester, and jumpstart.

The machine called admin runs the `monitord` process, which checks to see whether the processes that should be running on each of the services machines ARE running on those machines. If not, it restarts those processes.

admin also serves in a variety of administrative roles, including software distribution.

admin is an Ultra-170 with 192 MB of RAM, the usual two internal disks, and a 10-Mb Ethernet card. In addition, in the current implementation, it connects to a 56K leased line for credit card verification.

The harvester machine copies system logs from each of the services machines and runs analytical programs to distill the relevant information from the logs, storing a summary and producing a variety of graphs. In particular, data from these reports can pinpoint locations where local Internet providers have not met demand from WebTV's customers.

The current harvester supports both the Alma and Bryant sites; it is an Ultra-2 with the maximum allowable RAM: one Gigabyte. Even at this configuration, it still takes twelve hours to process one day's logs. It does not seem to be memory-bound, and it may even be possible to cut back on RAM in this machine.

The disk demands by harvester are severe: a Model 112 SparcStorage Array has been ordered for it, with thirty 2-GB disks, the maximum the SSA can support. The Operations team suggests that two SSAs may be needed if a single harvester machine must support 50-60 services machines.

Because the harvester machine needs to get logs from all of the services machines, it needs two 10-Mb Ethernet interfaces to connect it to the iapisp and admin networks.

The jumpstart machine is used to load, reload, or update the operating system configuration on any of the services machines. An Ultra-170 is more than sufficient for this purpose; it needs a 10-Mb Ethernet card in order to be attached to the network of the machine being configured.

Database Machines

The database machines are used for two purposes: customer records and service information. A machine called custmaster stores customer information, including billing data, with custrecover serving as its backup. A separate service database is stored on custservice; an application server called remedy (named after the software it runs) is its front end.

Each of custmaster, custrecover, and custservice should be Ultra-2s. Each of the three needs three disk racks with eight 2-GB disks and dual power supplies in each rack. (See Chapter 4, “Configuring the Systems and Software” for the uses of these disk racks.) All have four SunSwift SBus controllers, one for the internal disks and three for the disk racks.

The machines custmaster and custrecover need 1 GB of RAM to run large Oracle applications; custservice runs with 192 MB of RAM.

The Remedy application server, called remedy, is an Ultra-170 with 192 MB of RAM. In the current configuration, there is also an Ultra-170 which is used by the DBA to collect statistics from the other machines.

Each of the database machines needs a 10-Mb Ethernet card to talk to the db network. (remedy needs one, but its location is usual with the support organization, not on the service networks.) If there is a db-admin network then the three database hosts will also require 100-Mb FastEthernet interface cards.

File Servers

File servers are used to store persistent information which might need to be accessible to any of a number of service machines. The principal information stored on WebTV file servers is electronic mail, user favorite-site lists, and "cookies".

"Cookies" are user-specific information that particular Web sites expect the user to store. They are often used for security purposes, to control access to parts of a site. A common cookie implementation is an encrypted or random string that can be queried when the user attempts to access a restricted page.

Cookies are also used to gather information for advertisers about behavior, because each cookie is unique to its user.

WebTV uses NFS file servers from Network Appliance, Inc., colloquially referred to as NetApp. Their file servers' reliability and performance are considered extremely high, and most reviewers have concluded that their price/performance is superior to any offered by NetApp's competitors.

The particular model used is NetApp's high-end F540, which supports up to 200 GB of disk on a single machine. WebTV purchases this machine with 8 MB of NVRAM, and 100 GB of disk space on two Fast-Wide differential SCSI controllers. The size of this machine allows all user electronic mail to be stored on a single host, making NFS-mount configurations easy to administer.

The F540s need a 100-Mb FastEthernet interface to the store or newsstore net, and a 10-Mb Ethernet interface to the admin net; for these connections, the configuration includes two 10/100 auto-sensing Ethernet controllers.

For fast repair, WebTV also recommends the purchase of the F540 Spares Kit with at least one of the file servers.

One of the Alma deployments of the NetApps dedicated one file server for mail storage, and one for all other storage. The latter machine has proven to be much more heavily loaded than the former. Removing harvester's demands on the file servers has balanced the space demands on the two file servers, but the "otherstore" server still has twice the processing load of "mailstore".

One solution proposed is that cookies should be cached by the proxy servers, which currently have to retrieve them repeatedly when a user is visiting a sequence of related Web pages.

Instead, the Bryant implementation has merged the two functions onto a single NetApp, with the intention of creating a new "service group"—with its own NetApp—when the first one approaches capacity. (A "service group" is a set of WebTV customers using the same implementation of the service; service groups are created either to test or run different versions of the software, or to redirect them to a new hardware implementation of the service.)

Tape Drives

The only services machines which are backed up are the NetApps and the active database machines. It is assumed that the other machines hold only transient data, along with operating systems and service software which can be reloaded.

One backup unit is purchased for each NetApp, and for custrecover. (custmaster and custservice copy their data to custrecover, which is then backed up to tape.)

The backup units deployed are Quantum DLT 4700 tape stackers. Each stacker comes with a single magazine, seven DLT tapes (at approximately $100 per tape), and a cleaning tape. With each unit, an extra magazine and seven extra tapes are purchased.

The tape device directly connects to a SCSI port on the NetApp, but the device needs 50-pin Centronics SCSI and the NetApps provide a Sun Mini-D50 SCSI connector. Thus, a connector cable must be purchased with the unit.

The tape drives use DLT technology, 20 GB of data can be stored on each tape before compression; 40 GB compressed. Thus, each stacker can store 280 GB. Quantum is the only manufacturer of this technology; WebTV's suppliers have been R Squared, Vanguard, and Acclaim, depending solely on price and availability.

The 4500 model--which holds five tapes rather than seven--may be sufficient, but has been hard to find. Future models may store even more data per tape, and should be considered when available.

No more tapes are currently needed, because at this point backups are performed only for disaster recovery. No commitment has been made to customers that individual data, including electronic mail and favorites lists, will be available for restores. Thus, only whole file systems need to backed up, and older copies do not need to be retained.

Sharing tape drives between machines has been proposed, but the NetApp remote tape backup utility was writing a gigabyte of data in 33 minutes. As a result, each NetApp has its own stacker.

The Number of Machines Needed for Each Machine Class

The art of building a WebTV service implementation is in selecting the number of servers needed for each class of machine. The results do not have to be perfect: configuration files can be adjusted to redistribute processes from one machine or class of machines to another.

That said, there are almost certainly some rules of thumb to follow when making initial purchases. (Among them: There are more mailtods than postoffice machines.)

It is suggested that two hundred users can be served by two proxy machines, and one each of the mailsorter, mailtod, storefull, storeless, harvester, and NetApp machines. Other functions can probably be distributed among these as experience dictates. In this configuration, it would probably be appropriate to try combining the iapisp and store networks, probably by using a dedicated 10Mb switch.

This section should be expanded to include many more of these rules of thumb.

Internet Providers

WebTV uses a distinction between Internet Services Providers (ISPs) and Internet Access Providers (IAPs). IAPs allow a WebTV customer's device to connect to a WebTV service; ISPs are used by the service to respond to requests for information retrieval and mail delivery.

Internet Service Providers

WebTV uses two different ISPs in its Palo Alto, California, service locations: BBN Planet and UUNet Technologies. These two providers have extensive, high bandwidth backbones, and are connected in multiple locations to other providers; these characteristics minimize the time needed to respond to a request.

Right now, there is a single DS-3 connection coming from BBN into the service on Bryant, and one coming from UUNet into the service on Alma. The two services are connected by a DS-3 so that each can serve as the others' backup in case of outage or overload. Only about 9 Megabits/second of the theoretical 45 Mbps capacity are currently used, even at the peak demand time of 7:00 p.m. Pacific time. (The peak demand time could shift as days grow longer, and is also probably affected by the location of the service on the West Coast.)

There is some discussion taking place about whether smaller connections to other ISPs would improve performance enough to justify the additional expense.

Internet Access Providers

Internet Access Providers are chosen for their local presence in a large number of locations, or in locations not served by WebTV's existing IAPs. CNC and Ziplink are two of WebTV's major IAPs, and UUNet serves as an IAP as well as an ISP.

There is one DS-3 from CNC into each of the Bryant and Alma facilities, and one from Ziplink into Bryant. The CNC links seem to peak at 8 Mbps and 2 Mbps in Bryant and Alma, respectively. This imbalance is due to a deliberate trend to redirect connections to the more spacious and sophisticated machine room at Bryant.

Where possible, an IAP serves a dual function as an ISP. WebTV configures its routers to let a query bound for a site hosted at a particular provider go directly through that provider. Both CNC and Ziplink serve currently as smaller ISPs when such a use would shorten the number of hops between WebTV and the host being queried.

Domain Name Registration

New services may possibly require new domain names. If so, those names would have to registered in time for the services to use them.

Internet Router(s)

For the connections to the Internet providers, WebTV now recommends a Cisco 7206 with IP-only software, an interdomain routing license so it can run BGP, 64 MB of RAM, and 16 MB of flash memory. (PCMCIA flash memory "credit cards" come in sizes of 8, 16, and 64 MB--16 is right.) The router should have 2 Fast Ethernet ports and whatever interfaces the Internet providers specify. (As an example, in Fujitsu Trial 1, WebTV will be the sole IAP/ISP for the trial, so a single additional Fast Ethernet port will suffice. More typically, two Internet providers

might each ask for a single HSSI port.) If possible, a single router should have two power supplies for increased reliability; if more than one router is deployed, dual power supplies in any one are probably not needed.

In the past, a Cisco 7507 has been used in this role. It would be configured with IP-only software, an interdomain routing license, a VIP2-20 card with two FastEthernet port adapters, additional VIP-2 cards with the appropriate interface port adapters are needed for the IAP/ISPs, 64MB RAM, 16 MB flash, and 2 power supplies.

WebTV recommends this vendor and router for the following reasons:

1. Cisco has the best Internet-connected routers on the market.

   Wellfleet makes decent routers, but the predominant Internet router vendor is Cisco. The company provides good support, leads the market, is extremely solid financially, participates actively in industry standards committees, and its market position assures connection compatibility with existing Internet Service Providers.

2. BGP (Border Gateway Protocol) is the preferred protocol for connecting multiple Internet providers.

3. The hardware configuration is driven by the need to connect the iapisp and router-nexus internal networks with two external networks, one for each of two Internet providers. The RAM is needed for BGP; at least 32MB are needed, and 64 MB is much safer. The 16 MB flash memory requirement is recommended because Cisco software images keep growing, and the current image is already 4-5 MB. Good administration practice recommends the ability to preload a second image of approximately the same size, so 8 MB is not sufficient.

4. The 7500 line is Cisco's high-end line. The 7000 is older technology, and probably won't be sold by 1998. The 7500s have higher performance than the 7200s, but the 7500 is approximately $30,000 more expensive than the 7200, which costs only about $40,000 in the configuration needed. Even if every port in these configurations was operating at full speed, a 7200 would be sufficient to meet the bandwidth demand. It can transport 600Mb/second in theory, though perhaps only 500Mb/second in practice. Given that two FastEthernet ports and two HSSI ports--used to support two Internet connections--can require a maximum of only 290Mb/second at single duplex, and thus a theoretical maximum of 580 Mb/second at full duplex, the 7200 models will almost certainly be sufficient.

   Even if we want to grow beyond these speeds, the preferred approach to adding bandwidth would be to add a second chassis rather than additional ports on a 7500. The second chassis would add reliability as well as capacity--an advantage over adding ports to an existing chassis. The backplane of a single router would no longer be a single point of failure, and a second router chassis would make it easy to take down one router for maintenance while continuing to provide WebTV service.

   Only in the case where it was known that there would be great demand for bandwidth at a single site, would it be preferable to use a pair of Cisco 7507s, or even 7513s or higher, for the pair of routers.

   There is the possibility that the 7200s might not turn out to be the right choice, but even in that case the ports, flash card and perhaps even the license--

approximately 30% of cost of the 7206—could be moved into any future 7500, or the 7200 could be used on the internal networks in the company.

Network Device Menu

Each of the networks used in the WebTV service is implemented with a network device, or a set of related devices.

There are three different network solutions used in the service: shared 100Mb FastEthernet, in which a number of hosts compete if necessary for a high-speed connection; dedicated 10Mb Ethernet, in which each host has a slower link all to itself; and dedicated 100 Mb FastEthernet, in which each host has a dedicated fast link.

For the first of these, a Cisco 116T+ FastEthernet hub is used, purchased with its optional network management module. Each host on the network requires a port on a hub, so if the number of hosts exceeds the 16 ports on this box, multiple boxes must be purchased. This Cisco solution is preferred over the otherwise acceptable 3Com solution, because it is “a class 2 repeater rather than class 1." As such, multiple hubs on the same network can be connected with up to a 20-meter long 10BaseT cable instead of having to use a 12-inch proprietary bus extender between them.

These hubs can be expanded to support up to 32 ports, but having that many hosts share a collision domain is viewed by the WebTV network folks as a potentially problematic solution. The cost for the 116T+s is approximately $3-4,000.

When dedicated 10Mb is needed, a Cisco 1900 Ethernet switch is used; these boxes have no additional options or pieces. They are very economical ($3K for 25 10Mb ports and 2 100Mb ports), and meet the technical requirements, including “full bridge functionality”--the broadcast of a packet to all ports when its correct destination is not yet known.

WebTV tried 3Com switches but had problems because they didn't implement full bridge behavior: If a packet is sent to a host that shares a switch with the sender, and whose connection has timed out, the 3Com will send the packet only to the designated uplink port, rather than sending it to all the ports on the box. Thus, the packet is never delivered.

Cisco 1900s are in the same price range as the 3Com alternative, have better functionality, and occupy less rack space.

WebTV uses only 24 of the 25 ports available on the 1900, leaving the inconvenient 25th--which is on the back of the switch and needs a transceiver because it's an AUI--as an emergency expansion port.

Finally, WebTV also uses the Cisco 2900--a 100Mb switch for dedicated 100Mb speed. This device is used to aggregate switched 100Mb hubs and/or dedicated 10Mb switches. The 2900 can then be connected to the 100Mb ports on any 1900, 116T+, or router. All of its ports can be full duplex 100Mb, but it should be remembered that the 116T+'s can only take half duplex, because of the protocol specifications for FastEthernet hubs. A Cisco 2900 costs around $10,000 and has 14 ports.

All of the Cisco devices are price-competitive, and because the vendor for hubs, routers, and switches is the same, WebTV gains the advantages of having a single supplier--easier vendor management, volume discounts, and single point of contact for support.

One caution, however, on purchasing from Cisco: It generally takes four weeks to receive any kind of Cisco equipment AFTER the PO is placed.

The Devices on Each Network

1. The iapisp net is dedicated 10Mb, a choice supported by WebTV's theoretical performance models. Load is distributed fairly evenly among the various servers dedicated to each process, with round robin assignment of connections by the headwaiterd process.

   Given that the total bandwidth to the Internet providers is currently 90 Mb/sec, there seems little to be gained by giving 100 Mb/sec connections to each of many individual servers. The ability to burst to 100Mb speed, therefore wouldn't actually provide much advantage, and the guaranteed 10Mb speed provides assured packet flow for each server. The Bryant Street implementation is running successfully this way, and most of the Operations team believes this approach is to be preferred.

   Another benefit to the 10Mb/sec choice is that the 1900's take up less rack space than other network devices, they provide more ports, and they are also used as internal hubs, gaining an economy of scale in purchasing.

   For all these reasons, the iapisp network is a dedicated 10Mb net. Having determined the number of servers on this net, one need only order the number of 1900's needed to provide a port for each. If more than one 1900 is needed, one or more 2900's should be ordered to connect them together.

   To date, there has been no need for more than one 2900 on a single network, but if needed, two of them could be connected by a 100Mb cross-over cable. If they are used to build the iapisp network--the most likely application of multiple 2900's-- AND there are two routers, then one router should be connected on each 2900.

   Solutions for expansion beyond these configurations have yet to be determined, and may prove unnecessary if multiple sites are built instead of expanding one large one. The Bryant street implementation has two 2900's, but only one is in use at this writing.

2. The store network is shared 100Mb network. It generally has one or two file servers doing large data transfers to a small number of proxy or mailtod machines at a time. Data transfers must be fast, but the number of simultaneous transfers is assumed to be relatively small. Transfers to the harvester NetAPP in particular are very large.

   The store network then is implemented with one or more 116T+'s. Enough of these hubs need to be purchased to support the number storefull, storeless, and NetAPP's with interfaces on the store net.

   At this writing, the Bryant site has two 116T+'s connected with a crossover cable. In the future, multiple hubs could be connected with a 2900.

3. The newsstore network is also a shared 100Mb net. Its file servers contain news articles to be distributed as requested. 116T+’s are used to implement it.

4. If there is only a single Internet router, then the router-nexus net can be implemented with a single 1900. A dedicated 10Mb port is used for the database firewall router, and one of its two 100Mb ports can be used for the Internet router.

   If a second Internet router is added, a 116T+ should be added, connecting into one of the 1900's 100Mb ports. Both of the Internet routers then plug into the 116T+. Deploying a 1900 here would require tying up both of its 100Mb ports with the Internet routers, leaving no port into which network debugging equipment like a sniffer could be plugged. This free port should also be used for pre-staging a new router.

5. The db-admin net, needed if a backup database server is used, should be shared 100Mb, because its few transactions are large file transfers. With only two machines, a cross-over cable is all that's needed. If more than two machines are used, a 116T+ should be ordered.

6. The admin network can be dedicated 10Mb, because administrative access does not require large bursty traffic. Thus, 1900s are used to implement this network. They have the advantage of low cost, small size, and functionality which could be used in the future to create different administrative access to different classes of machines.

   Enough 1900's should be ordered to connect all machines which need to be administered: those on the iapisp, store, and possible db networks. Given the number of machines, it is very likely that a 2900 will also have to be ordered to connect the multiple 1900's needed.

   If cost constraints are severe, the admin and db nets could both be shared 10Mb, using inexpensive hubs instead of 1900's. That would eliminate the possibility of taking advantage of the 1900's security option, and could conceivably be a performance problem for updates to the hosts. Jeff Jensen recommends instead that money could first be saved by eliminating the second power supply on the routers, and perhaps even eliminating the admin network altogether.

7. The modems are connected directly to the router-nexus net. They have 10Mb connections.

SCSI Controller Cards and SCSI Connectors

The SunSwift controller cards used by WebTV are the wider ones, chosen at one point because some machines needed not only 68-pin FastWide SCSI connectors, but also two 100Mb FastEthernet connections.

The SCSI connectors that have to be plugged into these controllers on the Ultra machines have turned out to be too wide to fit into the recessed mount point on the backs of the Ultra's. WebTV administrators are having to trim off some of the insulation on each cable in order to connect it.

Console Servers

All servers, hubs, switches, and routers have serial console ports for access by administrators. These ports could be connected to individual consoles, but those consoles would take up a large amount of space. Instead, the serial ports are connected through the pre-wired network cabling and patch panels to Remote Terminal Servers, or "console servers".

WebTV uses Cisco 2511's for its console servers, which have sixteen ports each to accept serial connections.

Cables and Connectors

Patch Cords and Console Cables

WebTV buys all of its patch cords and cables from Americable, which has decent pricing and stocks all the colors and models needed.

Both the console and network cables are 4-pair, Category 5, straight-through EIA568B cables terminated on RJ-45 jacks on both ends. Both ends of these cables have boots and each has a serial number printed on it. Four different lengths are stocked: four-foot cables for short runs, ten-foot for medium runs like those up and down a single rack, and fifteen- and twenty-five-foot cables for longer runs between racks. Care is taken to use the right length for the task at hand: extra-long cables get messy, and extra-short cables can't be easily moved aside.

Each cable is manufactured with a single serial number; the same number is printed on each end so that an administrator tracing a cable can be assured of its identity. A cable serial number is of the form *pn-lng-nnnnn,* where *pn* is the two-digit WebTV part number, *lng* is the cable's length in feet, and *nnnnn* is a unique identifier for that cable. This last number is unique within a single part number group, but may be reused in a different part number. (In other words, there may be both a 01-10-21634 cable and a 23-10-21634 cable.)

As an example, here are the part numbers currently in use at WebTV, and a snapshot of the serial number records that was taken on March 18, 1997. The records are kept to indicate which serial numbers will be used next in each series. Because these cables are used both for production and internal networks, not every part number is needed for a service buildout.

01 = 568B CAT-5 Patch Cable in WhiTe [NEXT S/N = 50186]

02 = 568B CAT-5 Patch Cable in BlacK [NEXT S/N = 50166]

03 = 568B CAT-5 Patch Cable in ReD [NEXT S/N = 50306]

04 = 568B CAT-5 Patch Cable in GreeN [NEXT S/N = 50251]

05 = 568B CAT-5 Patch Cable in YelloW [NEXT S/N = 50221]

06 = 568B CAT-5 Patch Cable in BLue [NEXT S/N = 50341]

07 = 568B CAT-5 Patch Cable in BROWN [currently unavailable]

08 = 568B CAT-5 Patch Cable in ORange [NEXT S/N = 50251]

09 = 568B CAT-5 Patch Cable in GraY [NEXT S/N = 50696]

10 = 568B CAT-5 Patch Cable in VIOLET [NEXT S/N = 50001]

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31 = 568B CAT-5 Patch Cable in PINK [NEXT S/N = 50001]

32 = 568B CAT-5 Patch Cable in BeiGe [NEXT S/N = 50351]

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11 = CDDI crossover is gray with RED no-snag boots [NEXT S/N = 50001]

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21 = 25pair CAT-4 FEEDER Cable (180Male to 90Female) [NEXT S/N = 50001]

22 = 25pair CAT-4 FEEDER Cable (90Male to 90Male) [NEXT S/N = 50001]

23 = 25pair CAT-4 FEEDER Cable (90Male to 90Female) [NEXT S/N = 50001]

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41 = FDDI ST - ST Patch Cable [NEXT S/N = 50011]

42 = FDDI ST - MIC Patch Cable [NEXT S/N = 50001]

43 = FDDI MIC - MIC Patch Cable [NEXT S/N = 50001]

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51 = 10BaseT CROSSOVER is black with GREEN no-snag boots [NEXT S/N = 50156]

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61 = ATM Cable (SONET/SDH multimode)[NEXT S/N = 50001]

62 = DS3 Cable (Coax RG59 Male/Male 75Ohm PVC)[NEXT S/N = 50019]

Each server, router, and modem bank needs one cable for every network switch or hub to which it will be connected, and additional cable for its console. The cables specified can serve as Ethernet, FastEthernet, and console cables. Each network switch or hub needs one cable for each device that will connect to it, including one for its console. Only a fairly detailed network diagram that accounts for all devices can specify exactly the number and lengths of cables needed; extra cables and extra length are to be preferred when doubt persists.

Console Adapters

Most pieces of computing equipment used in a WebTV service were designed to use a terminal or a PC as a console. WebTV saves space by connecting the console ports of its equipment to Cisco 2511 terminal servers. With this modification, however, comes the complication of connecting to the RJ-45 port on the 2511's. Most types of equipment require some type of adapter to complete this connection; this connector is called a console adapter.

For each Sun server, WebTV uses a single part as a console adapter; the Americable part number for this console adapter is 24238-DTE-BK.

For each Cisco 1900 switch and Cisco 116T+ hub, a DB9 console adapter is needed; the part number is 24245-DTE-BK.

For each Cisco 7200 series router, a single DB25 console connector is used.

Of the equipment used at WebTV, only the Cisco 2511's and 2514's don't need a console adapter. (The connectors which come with the 2500 series devices, however, are not used by WebTV.)

Specialty Connectors

For each Cisco 2511 terminal server, an ATI AT-MX20T-04 AUI transceiver must be provided in order to connect it to the Ethernet admin network.

To gain an additional FastEthernet port on each 7206, its MII FastEthernet port must be converted to RJ-45; a Canary CTX-1070 is used for this purpose.

For the smallest implementations, a cross-over cable can substitute for a hub or switch that would otherwise connect to only one host.

Telecommunications Access

Though WebTV customers connect to the service primarily through Internet service providers, there are two circumstances which require connection through the public switched telephone network.

The first case is upon initial connection, or after the client box is moved. The client box dials a toll-free number, which is answered by a bank of modems in WebTV's Palo Alto office. The line, called the “preregistration line”, is a single PRI connected to a bank of modems. PRI technology is used because it is more reliable analog solutions in providing Automatic Number Identification.

The single PRI, supporting up to 23 simultaneous callers, has proven more than sufficient for the preregistration application, experiencing no problems even during the Christmas rush of more than 4000 new customers in a single day. (An average transaction on this line takes 22 seconds; few calls are over 30 seconds long.)

The second application is more demanding of resources; it provides a toll-free connection to the service when the customer's local Internet provider doesn't answer. These calls can last for hours-- the length of the customer's session. Some calls have been known to last for sixteen hours.

Multiple PRIs are used to handle this traffic, configured by the provider in a hunt-group. With 35,000 customers, five PRI's are used and even that number is not enough; for two hours each weekday evening, the lines take new calls as soon as they release old ones.

WebTV's provider had to provision a multiple-PRI hunt-group in a specialized location, using a DMS-250 phone switch and a feature called “Office Routing.” The alternative to using multiple PRI's would be to employ a swarm of analog lines; the wiring complexity with analog makes it a very distant second choice.

Modems

The modems that answer these calls are housed in a rack from US Robotics. The model is the Total Control 16, configured with:

- one dual-PRI interface card, software version 2.0.3 or higher, and
- one PRI NetServer Gateway card, version 3.2.27 or above, and
- twelve quad digital double-sided modem ATI7 cards, version 3.5.6 or above. (V.34 bis.) Don’t run version 4.1.1 of the modem software, because it scrambles the ANI if the rack is powered down.

The network management card in the Total Control 16 is running version 4.1.3.

The rack provides seventeen slots and all its configurations include a network management card. The NetServer Gateway card serves as the rack's 10-Mb interface to the router-nexus network.

The two PRIs and forty-eight usable modems in a single rack support up to forty-six simultaneous users. With five PRI's dedicated to backup service and one to preregistration, WebTV currently has three USR Total Control 16's.

Caution: USR leadtimes were 6-8 weeks before last year's winter holidays, although two weeks is the more common experience.

Software

There are a number of third-party pieces of software that are used by the WebTV service. This section should be expanded to describe their uses and version numbers.

Oracle is run on the database servers; the version we're running is Oracle 7.2.2.3.

Most queries which request auditory output are routed directly from the client to the Web server on the network, bypassing the WebTV service. However, some of WebTV's own content produces sound, so at least one of the service servers runs a realaudio server. The license charges for realaudio may be included in the deal for the WebTV service. This paragraph needs to be expanded with details, or dropped if realaudio is included.

Similarly, the filtering programs, currently Surfwatch and Yahooligans, may also need additional licensing arrangements, or replacement in other implementations.

Chapter 3

Hardware Installation

This chapter describes how to perform the physical hardware installation and cabling of a WebTV service. It is in an extremely immature state and will be expanded in future revisions. The following topics are covered:

- Facilities, Power and Cooling
- Racks, Patch Panels, and Wiring Runs
- Terminal Servers
- Network Devices
- Telecommunications Installation
- Internet Routers
- Internal Routers and Database Network Gear
- Modem Racks
- Support Servers
- Service Servers
- Connecting Computers: General Rules
- Database Servers
- File Servers

Facilities, Power and Cooling

WebTV services run in a computing environment which helps to guarantee its near-continuous availability to the customers it supports. Sufficient space is provided to install the racks already purchased.

Power

All WebTV equipment runs on 120-Volt power. Three-phase electrical power should be run overhead to the outlets on each rack, with each outlet on a separate leg.

Utility power is backed up by two Universal Power Supplies connected in series, each sufficient to handle the entire load of the service equipment. Behind the UPS's should be a generator capable of supporting the load indefinitely so long as it is refueled.

Cooling

The Bryant Street facility is currently using two 15-ton Liebert cooling systems, which are more than adequate for the load. A raised floor is not used, though one might provide even better air flow if it were installed.

Telecommunications Installation

WebTV uses two local telecommunications providers in its installations wherever possible. In the Bryant Street installation, the company found it advantageous to install fifty amps of 48-Volt power with redundant rectifiers to power the network cabinets from the two providers; this choice saved rack space in the machine room.

Racks, Patch Panels, and Wiring Runs

Most of the wiring in a WebTV service is installed even before any equipment arrives. A 24-port RJ-45 Category-5 patch panel is installed at the top of each rack. The wiring from this panel is run to two racks of centralized patch panels; these racks also house the network switches and hubs.

Terminal Servers

The Cisco 2511 terminal servers purchased to be console servers should be mounted just below the patch panels of the racks. There may be more racks than terminal servers; the equipment should be distributed at approximately even intervals to minimize the length of console cable wiring runs.

Each 2511 should use another 2511 as its console server, so DB-9 connectors should be run from each 2511's console port to one of the serial ports on another 2511. No console adapter is needed to connect these, but the connectors packaged with them are not used by WebTV.

Network Devices

The switches, hubs, and routers should be installed in the one or two central networking racks. Each one must be connected to a terminal server port using the appropriate length straight-through 8-wire (4-pair) patch cable. For each Cisco 1900 switch and Cisco 116T+ hub, a DB9 console adapter is also used to complete the console connection.

The admin network 1900's will be the most numerous; they can be mounted, connected to terminal servers, and daisy-chained to each other. (If a 2900 is ever used to connect the 1900's, it should be mounted first.) Because of the daisy-chaining, it's probably best to mount all of the admin 1900's close to each other, leaving room after them for future expansion.

For each Cisco 2511 terminal server, an ATI AT-MX20T-04 AUI transceiver and patch cable must be used to connect it to the Ethernet admin network.

The 1900's for the iapisp network should also be grouped together in the network racks, because they too will usually be daisy-chained together.

The 116T+'s which make up the store network should also be grouped together. With two 116T+'s, they can be connected with a cross-over cable. If there are more than three 116T+'s for the store net, the Cisco 2900 that connects them should be installed and the connections made.

The newsstore network device should be similarly installed.

The router-nexus switch or hub should also be installed, in a position which will allow convenient connections to the USR Total Control 16's, the Internet routers, and the database and admin routers.

Internet Routers

The Cisco 7206 routers used to connect to the Internet should be mounted at this stage, and connected to the iapisp network. If there is only a single router, and the router-nexus switch is a 1900, the router should be plugged into the 1900's single 100-Mb port. Otherwise the router(s) can be connected into any of the 100-Mb ports on the router-nexus device.

A console server connection should also be made; the 7206's need a single DB25 console adapter. To gain an additional FastEthernet port on each 7206, its MII FastEthernet port must be converted to RJ-45; a Canary CTX-1070 should be connected for this purpose.

The Internet access connections should be attached to the routers as they are installed.

Internal Routers and Database Network Gear

Two other routers, both Cisco 2514's, are sometimes used in building a production WebTV service. One, the database router, can now be mounted; it is probably preferable to mount it in the rack that will house the database servers. It should be connected to a port on the router-nexus device which supports 10Mb Ethernet.

If there is an admin router, it can also be mounted at this time, and connected to the internal network of the support organization.

The 2514's do not need special console adapters; they can be connected to the terminal servers through the standard patch cords.

If there are db (database) and/or db-admin network switches, they should be mounted in the rack that holds the database router. The database router should be plugged into the db net switch.

Modem Racks

The USR Total Control modem racks should be mounted and connected to the router-nexus device and to console servers. They should also be connected to the PRIs from the telecommunications provider that is supplying the toll-free dial-in service.

Connecting Computers: General Rules

Virtually all of the devices that remain to be connected are server machines. Connecting one of these machines will have four parts:

1. Mounting the machine in a rack with enough space to support its peripherals,
2. Connecting the machine's serial port to a terminal server,
3. Connecting the machine to the admin network, and
4. Connecting the machine to any other networks its function requires it to access.

All of the Sun servers--the service servers, support servers, and database servers--make their connections to console and to the admin network in the same way. The console connection requires a console adapter to convert its DB-25 serial port to RJ-45; the Americable part number for this console adapter is 24238-DTE-BK. The admin network connection is made from a 10-Mb interface card on the Sun using a patch cable.

In many cases, the admin network connection is made first by running cables up to the patch panel at the top of the rack housing the machine. The connection is then completed on the network rack by connecting the corresponding patch-panel port to an admin switch.

This option is available, if necessary, for connections to console servers and to the other networks.

Support Servers

The admin server needs no more than the standard connections, with one exception. In the Bryant Street buildout, it is also connected to a 56K leased line which is used to verify credit card balances and validity.

If there is a separate jumpstart server, it can use just the standard connections. One implementation even disconnects it from all networks when it is not in active use.

The harvester server must be mounted with its Sparc Storage Array. In the past, it has been connected to the iapisp net as well as the admin net. (If it uses a NetApp file server on the store network, it should also be connected to the store net.)

Service Servers

Proxy servers must also be mounted with their disk racks, and their three SCSI controllers connected to that rack. The proxy servers are connected to the iapisp net as well as the admin net. The mailsorter machines are handled almost identically, except that their three SCSI chains are each two disks long instead of three.

The postoffice machines are installed in pairs, along with their shared disk rack. Each needs a connection made from its single controller card to two of the four disks in the shared rack. They are then connected to the iapisp, admin, and store networks.

Mailtodr machines are handled in the same way as postoffice machines: they come in pairs, the pairs share a disk rack, and they connect to the same networks as do the postoffices.

Both the postoffice and mailtod machines share a similar problem, as well: the SCSI connectors to the SunSwift controller cards don't fit into the backs of the Ultras. The connectors are extremely wide; they have both 68-pin FastWide SCSI and 100Mb Enet connectors. As a result, it is necessary to trim some of the insulation off these connectors so that they can make clean connections to the recessed port on the Ultras.

These connectors are often used on other servers as well, but future configurations may take advantage of the fact that not many of the machines need connections to both a 100-Mb network and a FastWide SCSI device.

The storefull machines do not have external disks, but need to be connected to the admin, iapisp, and store networks. Store network connections for these machines--and for all machines on that net--are from 100Mb interface cards. The others are 10Mb cards.

The other services machines need only be connected to the admin and iapisp networks, except perhaps the various news machines. Information about the latter will appear in a later revision.

Database Servers

The database server(s) need a connection to both the db and db-admin network switches, if those switches exist. If neither exists, there must be only one database server; that server should be connected with a cross-over cable to the database router.

A DLT 4700 tape drive should be connected to the database server called custrecover-1, or to the single database server if there is only one. The DLT 4700's are not rack-mountable and they're very deep, so they overlap the end of the racks' shelves. Both the Suns and the DLTs should probably be attached for earthquake-proofing.

File Servers

The Network Appliance file servers are the final machines that need mounting. A DLT 4700 tape drive connects directly to a SCSI port on each file server; a conversion cable must be used to connect the tape drive's Centronics SCSI to the Sun Mini-D50 SCSI used by the NetApps.

Finally, the NetApp's two 10/100 autosensing Ethernet interfaces should be connected to the admin and store (or newsstore) networks.

Chapter 4

Configuring the Systems and Software

After a piece of equipment is installed, it needs to be properly configured. Software needs to be loaded, system variables set, and initialization commands given. Finally, the software which runs the database and the WebTV service needs to be installed on the appropriate servers. This chapter describes these configuration and software installation procedures.

Note: This chapter is currently in an extremely preliminary state; it will be corrected and expanded in future editions of this book. The following topics are covered:

- IP Addresses
- Establishing Console Service
- Network Equipment
- Modems
- Internet Routers
- Jumpstart Server
- Database Servers
- File Servers and Tape Drives
- Harvester Server
- Service Servers
- Database Servers
- Database Software
- WebTV Software
- Custom Software
- Support Software

IP Addresses

To configure the equipment, two things are needed besides the hardware itself: a dumb terminal or PC to serve as the initial console, and the IP addresses to be used in the service. In previous installations, these have been provided by WebTV.

The iapisp and nexus networks have to have real IP addresses because machines on these networks must be able to make connections to other machines on the Internet. They must be routable, traceable through *traceroute*, and registered to WebTV.

Similarly, true IP addresses are needed for the PPP connections made into the toll-free number when clients can't connect to local Internet points-of-presence. These addresses, of course, can be reused repeatedly as connections come and go.

The pre-registration service does not need to use routable IP addresses, since the traffic is between the client and the service only. For this purpose, unroutable 10.x.x.x addresses are used.

The store, newsstore, admin, database, and db-admin networks don't need real IP addresses because their hosts are not to be seen outside the service. The IP addresses on these net aren't routable. The 172.16.x addresses are used; they belong to the set of addresses reserved for unroutable networks.

Establishing Console Service

The computers and network equipment that support the service are all purchased without a terminal, keyboard, or monitor. To configure these devices, some kind of console service must be established and connected to their serial ports. The first major step in configuring a WebTV service is to get its console server running on the machine destined to be its admin server.

Terminal Servers

Any dumb terminal, or a PC, should be temporarily connected to the serial port of each of the Cisco 2511s in turn; after configuration, the 2511s will be the terminal servers for the rest of the equipment (and for each other). Their ports are configured in *enable* mode. Once in enable mode, the following commands should then be given:

```
line 1 16

no exec

transport input telnet

flowcontrol hardware
```

At this time, the IP address for the 2511 should also be set, and the rest of its configuration completed.

The Admin Network Switches

The switches that link the machines on the admin network also link all of the 2511 terminal servers to the admin server. These admin switches should now be turned on. Their default configurations will be sufficient for the configuration process.

The Admin Server and Console Service

The admin server runs the console server and the process *monitord* that monitors the other WebTV software processes. In smaller installations, it is to be used as the jumpstart server for the other support servers and all of the service servers.

The admin server can be given its final configuration at this point if that configuration is available on tape, or can be transmitted over a network connection. If not, the machine need merely be configured with a version of the Solaris 2.x operating system that is capable of loading, configuring and running the console server software.

WebTV runs a customized version of the publicly available console server software; it should now be loaded on the admin machine. Its conserver.cf file should then be edited to the proper configuration.

Internet Routers

The Internet routers' configurations should include the following lines:

no service udp-small-servers

no service tcp-small-servers

ip subnet-zero

no ip source-route

The configuration should also turn on SNMP and turn off telnet access.

Network Equipment

The network switches and hubs can be powered up at this stage. The Cisco 2514 database router should be configured with the appropriate IP addresses for its interfaces, and its filters set to prevent unauthorized access to the database servers.

If there is an admin router, it can also be configured at this time, and the admin switch can also be configured to prevent traffic from the externally visible networks from crossing into the internal ones.

Modems

The US Robotics Total Control 16 modem racks also need to be configured. In the WebTV environment, each one's NetServer Gateway card serves as the rack's 10-Mb interface to the router-nexus network. Here is the file used by Jim Kubon to guide his configuration of that card:

Warning verify that the netserver card is running version 3.2.27 comOS

(issue the "version" command at the command prompt.)

If not upgrade to at least version 3.2.27 using pcsdl or Total Control.

Setup

set net0 address <IP Address> # set the IP address

set net0 netmask 255.255.255.0 # set the netmask

set net0 broadcast high # set broadcast address to xx.xx.xx.255

set gateway <xxx.xxx.xxx.254>

set password <password># sets !root password

set !rootaccess off # disable !root logins over dialup lines

set namesvc dns # sets nameservice to DNS

set nameserver <IP Address> # sets primary nameserver IP address

set nameserver 2 <IP Address># sets secondary nameserver IP address

set domain < domain name> # set domain name

set authenticate <IP Address> # sets primary RADIUS servers IP address

set alternate <IP Address> # sets secondary RADIUS servers IP address

set accounting < IP address> # sets primary RADIUS accounting server's IP addr.

set accounting 2 < IP address> # sets secondary RADIUS accounting server's IP addr.

set secret < secret key> # set secret, must correspond to secret in RADIUS clients file.

set all login network dialin # sets all dialin ports to login/network dialin

set pppmodem on # sets all modems up for PPP

set modem s1-s46 active

set all security on # sets up netserver to use RADIUS authentication

set all message WebTV Networks Inc. # sets login message to WebTV Networks Inc.

set all idletime 5 # sets idletimeout to 5 minutes

set sysname <system name> # set the name of the netserver

set prompt <prompt name> # set the command prompt to whatever

save all # save everything you just entered

reboot # make changes take effect.

Jim also produced this display of the modem registers in the rack:

USRobotics Digital Quad HST Dual Standard V.34+ Fax Settings...

B0 C1 E0 F1 Q1 V1 X7

BAUD=115200 PARITY=N WORDLEN=8 DTE=GATEWAY NAC

DIAL=HUNT ON HOOK TIMER LINE=ISDN PRI

&A3 &B1 &C1 &D2 &G0 &H1 &I0 &K1 &L0 &M4 &N0

&P0 &R2 &S0 &T4 &X0 &Y1 %N6

S00=001 S01=000 S02=043 S03=013 S04=010 S05=008 S06=002 S07=060

S08=002 S09=006 S10=007 S11=070 S12=050 S13=000 S14=000 S15=000

S16=000 S17=000 S18=000 S19=000 S20=000 S21=010 S22=017 S23=019

S24=150 S25=005 S26=001 S27=000 S28=008 S29=020 S30=000 S31=000

S32=009 S33=000 S34=000 S35=000 S36=000 S37=000 S38=000 S39=013

S40=000 S41=000 S42=126 S43=200 S44=015 S45=000 S46=255 S47=034

S48=000 S49=016 S50=100 S51=064 S52=005 S53=000 S54=064 S55=000

S56=000 S57=000 S58=000 S59=000 S60=000 S61=000 S62=011 S63=011

S64=000 S65=000 S66=000 S67=000 S68=000 S69=000 S70=000 S71=001

S72=000 S73=000

LAST DIALED #:

LAST DNIS #: LAST ANI #:

Jim says the easiest way to set these settings from the terminal interface is with the comand:

```
AT&F1%D2S39=13S51.6=0S62=11S63=11S47=6&W
```

Jumpstart Server

Because most of the WebTV server machines are Sun equipment running the Solaris 2.x operating system, they can be configured from a single machine running *jumpstart*. Jumpstart allows a system administrator to create an OS image for each of the machines to be configured, and then have that image loaded on the *target* machine when it boots using *install mode*. In addition, custom scripts can be created for each target machine, to be run before and after the OS image is loaded; these scripts can include formatting and partitioning of both internal and external disks. Similarly, jumpstart customizations are used to specify the interface numbers for various controller cards, drawing distinctions between machines with three and four SBus slots, for instance, and those which do and do not have FastEthernet or FastWide SCSI on the motherboard.

For each class of machine to be jumpstarted, a configuration directory tells the jumpstart software which parts of the operating system to install and which should be left out. Each configuration directory is effectively a record of what its machine class should be. Jumpstart installs the specified operating system components, customized files, specified patches, packages, and network

configurations, and sets customized variables. The jumpstart scripts sometimes must be customized for different WebTV service environments; for instance, a service with the iapisp network split into two networks needed different jumpstart scripts. Similarly, a "demo" configuration has been used for small test installations.

The admin server can be used as the jumpstart server, unless a very large service allows the two functions to be separated. The OS location should be /export/install-2.5.1; jumpstart is in /export/jumpstart-2.5.1.

Currently, the jumpstart tree and Solaris 2.5.1 OS image (2.5.1 is needed to support the Ultra-2's) are backed up to an internal WebTV machine called *ops*, in the file /export/ops/jumpstart/jumpstart.dump.gz. The file is over 500 MB gzipped. To uncompress and restore it, the command

```
gzcat jumpstart.dump.gz | restore if -
```

can be used.

The current machine classes are: proxy, storeless, storefull, mailtod, proxy, ns, notify, prereg, postoffice, mailsorter, realaudio, harvester, admin. The database and the various news handling machines will also eventually get their own images. The proxy configuration is the most generic, so new machine classes are created by specifying their differences from the proxy configuration directory.

Among the things all machines share is a production account with uid 100. This account is used to run software installation and update scripts.

One of the files that must be installed on all the server machines is /var/opt/oracle/tnsnames.ora. It contains information about the database to which each machine connects.

For the Fujitsu Trial 1 service, this file contained:

```
CMJPN1=
(DESCRIPTION =
        (ADDRESS_LIST =
                (ADDRESS =
                        (COMMUNITY = tcpcom.world)
                        (PROTOCOL = TCP)
                        (Host = lexus.japan.webtv.net)
                        (Port = 1526)
                )
        )
        (CONNECT_DATA =
                (SID = CMJPN1)
                (GLOBAL_NAME = CMJPN1.WORLD)
        )
)
```

File Servers and Tape Drives

This section will be expanded in future revisions.

Harvester Server

This section will be expanded in future revisions.

Service Servers

This section will be expanded in future revisions. Currently, it consists solely of notes concerning the naming of service servers, and name service itself.

Hostnames

WebTV service machines have hostnames based on their primary function. The general form of a hostname is *function-number.networkname.place*.webtv.net, where *function* is one of the machine classes, *number* is a unique sequence number, *networkname* is the name of the network the machine is connected to, and *place* is the geographic location of the service.

Examples of functions include proxy, ns, storefull, storeless, and prereg. Networkname examples include iap, store, admin, and db. Locations include alma and bryant. So far, the Alma Street service has used one- and two-digit numbers in the *number* position, and the Bryant Street service has used 101, 102, 103, etc. for each of the classes of machines. Thus, a example machine name would be proxy-101.iap.bryant.webtv.net.

This naming scheme allows a machine to have one name for each interface; there's also a shorthand name for each machine which omits the *networkname*.

Name Service

Each WebTV production service has two name servers: ns-1 and ns-2 in the Alma service, ns-101 and ns-102 in the Bryant service. ns-1 is the master for all four names servers; they cache names and get updates once every two hours. The name server software is currently bind 4.9.4-P1 (patch level 1); soon it will be upgraded to 4.9.5.

Database Servers

The database servers' operating system is Solaris 2.5.1, matching the other servers. Before the database software is installed, Veritas software is loaded on the database servers.The minimum Solaris configuration does not include the ANSI C++ library, which is needed for the ORACLE install; that library (libC.so.3) should be installed on each database server.

Database Software

In the production services, there are up to three database servers. The custmaster server holds the customer records, including the service to which each customer subscribes and the billing information for that customer. The custservice database holds information used by the Customer Care team to provide customer support; this database will probably run on a custservice machine in the future, but right now it is merged with the custmaster database. The custrecover server is the

backup to custmaster. There is also a Remedy server, which runs the problem tracking program for Customer Care; it needs access to the custservice database. The database software currently running on custmaster is Oracle 7.2.2.3.

Trial Software Installation

An installation of the database on the Fujitsu Trial 1.5 service used the following procedure. It used Oracle 7.3.2.1:

Preliminaries

- Create an Oracle user account
- Make a mount point and mount the Oracle software CD
- Choose an Instance Name of the following format: XXYYYN, where
  - XX is a DB Type, one of CM, CS, and
  - YY is an Instance type, one of DEV, PRD, DEM, and
  - N is a Sequence Number.

Oracle Install

The installation of the Oracle RDBMS Software was done with the default parameters.

- Log in as user oracle.
- Set environment variables. The profile file included the following lines:
  - PATH=$PATH:/usr/ccs/bin:/sur/local/bin:/tools/admin/bin:.
  - ORACLE_TERM=xsun5; export ORACLE_TERM
  - ORAENV_ASK=NO; export ORAENV_ASK
  - ORACLE_SID=CMDEM1; export ORACLE_SID # Set after install Only!!
- Install the Oracle 7.3.2.1 CD.
- Go to the install directory on the cdrom: cd <mount point>/oracle/orainst
- Start the installer in character mode: ./orainst /c
- Complete the install selecting the default values.
- Select "Install New Product".
- Select the default ORACLE_BASE, which is derived from the mount point.
- Select the default ORACLE_HOME: ORACLE_BASE/product/7.3.2
- Select NO to whether to create a database or database objects.
- Select relink executables.
- Select overwrite root.sh
- Select the following products:
  - Advanced Replication
  - Oracle Unix Installer
  - Distributed Database option
  - Parallel Query option
  - Server
  - SQL*Net
  - TCP/IP Protocol Adapter
  - Pro*C

  - PL/SQL
  - SQL*Plus
  - Say no to XA Library, Spatial Data, Context Option, Pro*Cobol, and Pro*ADA.
- Select dba (default) as the DBA group.
- Select an ORACLE_SID (instance); for purposes of this iteration we're using: CMDEM1
- Select default SYS and SYSTEM passwords

Oracle Upgrade

The Oracle release being used by WebTV is 7.3.2.3 so a patch upgrade is required. Install the Oracle 7.3.2.3.x CD next - this is a patch intended to add to an existing RDBMS installation. The procedure is the same as the one above, but this time Update is chosen.

- Log in as user oracle.
- Go to the install directory on the cdrom: cd <mount point>/oracle/orainst
- Start the installer in character mode: ./orainst /c
- Complete the install selecting the default values.
- Select "Add/Upgrade Software".
- Select the default ORACLE_BASE, which is derived from the mount point.
- Select the default ORACLE_HOME, which is ORACLE_BASE/product/7.3.2
- Select NO to whether to create a database or database objects.
- Select relink executables.
- Select overwrite root.sh
- Select the same products as before, where available.

Install WebTV Database

The final part of this procedure is considered the least mature. The database information is created from an export of the database, and the import command currently needs to be run several times in order to successfully generate the database instance.

- Move the init file for this instance, initCMDEM1.ora, to $ORACLE_HOME/DBS/.
- Run server manager to create database: svrmgrl lmode=y
  - connect internal
  - startup nomount
  - @create_cmdem1.sql (creates the instance CMDEM1)
  - @catalog (creates catalog/data dictionary objects - views, synonyms,etc)
  - @catproc (creates procedures necessary for database operations, so that import, among other things, will work)
  - exit
- Run import: imp system/manager full=y file=fullexp.dmp, where fullexp.dmp is the name of the export file)
- Kill it when it starts to fail
- Run server manager: svrmgrl lmode=y

  - connect internal
  - alter user system identified by manager; (or whatever password)
  - shutdown
  - startup (to bring new rollback segments on line so import will work)
  - exit
- Run import:
  - imp system/manager full=y file=fullexp.dmp
- Continue to kill, run server manager, and run import until the import succeeds.

This procedure is probably not complete. Later on, some of the configuration files had to be modified, including /var/opt/oracle/listener.ora and /var/opt/oracle/tnsnames..ora on the database server, and /var/opt/oracle/tnsnames.ora on the service servers. Init scripts were also installed, and the instance name changed.

Production Database Layout

The WebTV production database runs on a much more sophisticated server configuration than the simple database server used in the procedure above. The database server currently deployed is an Ultra-2 with four disk controllers, one for each of three nine-disk Adjile disk racks, and one for the internal disk. The internal disk holds the basic Solaris install and associated software and file systems, mirrored using Veritas software.

The database disk layout uses a modified Oracle Flexible Architecture, which is Oracle's recommended structure for storing the tree structure of a database. Each of the three controllers can write to the nine disks in its Adjile; in practice, eight are actually used. The disks under each controller have the following roles:

Table 0-1 RAID 0+1 Layout

Disk	Controller 1	Controller 2	Controller 3
1	Primary 1	Primary 2	Primary 3
2	Primary 1	Primary 2	Primary 3
3	Mirror 3	Mirror 1	Mirror 2
4	Mirror 3	Mirror 1	Mirror 2
5	Backup 2	Backup 3	Backup 1
6	Backup 2	Backup 3	Backup 1
7	/archive	/archive mirror	/export
8	/export	/oracle, /apps	/oracle, /apps mirror
9	spare	spare	spare

The raw partition on the Primary disks is called /ora17d, on the Mirror disks /ora17dm, and on the Backups disks /ora17b. Each pair of disks with the same name is striped together; writing is done with a three-way stripe across all three controllers to the Primary disks. The striping width is 64KB.

The Mirror disks are continuous mirrors of the correspondingly numbered Primaries; note that mirroring is done across controllers to maintain access to data should a single controller fail. The Backup disks contain the BVOLs, or backup volumes are mirrored copies of the correspondingly numbered Primaries, with the copies updated once a day.

The /oracle and /apps disk stores software; the /export disks are used for temporary storage during computation or administration.

The database server custrecover-1 is currently running as a warm standby machine. It also has three usable controllers. Once in the life of the database, the DBA copies the /archive raw partitions from custmaster-1 to the CMPRD1 volume on custrecover-1, and copies the custservice-1 raw partitions to custrecover-1's CSPRD1 volume. These spare /archive partitions can be used to hold copies of the /archive logs, and thus can be used to re-executed committed transactions. It is intended that these copies of the /archive logs will be updated once every fifteen minutes.

The Tablespace Layout for a system that is intended to hold up to 250,000 subscribers is in the table below:

Table 0-2

Use	Number	Size of Each
SYSTEM	1	250 MB
REDO	6	25 MB
TEMP	1	250 MB
RBS	1	250 MB
CTRL	3	1 MB
DATA/INDEX	48	250 MB

Each disk layout has one CONTROL area, allocated one megabyte of space. Two 25-MB areas hold REDO logs, and sixteen 250-MB partitions for data.

WebTV Software

In the production environment, WebTV service software is updated frequently--at least once a week and more if needed. The various service groups, one for each location and several others for various testing and development purposes, can be updated in a process.

The installation process is actually three separate processes, each of which can be performed independently of the others. One installs the software, one the WebTV-specific Web content, and one the flashrom images that are downloaded to the client boxes when they need changes. The software takes approximately 300 MB on disk, the content around 20 MB, and the flashrom images around 100 MB. Enough disk is should be available on each machine to store at least two versions of each part of the service.

All of the software can be, and usually is, updated on all of the servers in a service group at the same time. The user production owns the files, and is used to run the update process. There is a production account on each service server, and a configuration file, ~production/webtv/config/ production.solaris2.cf, includes one entry for each server to tell that server its service group and its role(s) in that group.

The admin-101 machine in Bryant is used to update the software; there has been some discussion of using a separate machine called 'dist', but the operations team has not yet found a separate machine necessary. At the Alma site, a machine called techultra--located on the non-service side of the admin-router--is used to perform these installations.

Using ssh

The scripts which do the multi-machine installs depend on a UNIX utility called *ssh*. The name stands for "secure shell." Like rsh, ssh allows a login to a machine without the need to provide a password each time. The user need merely type her passphrase once during a session, and the ssh agent will provide authentication to each authorized remote host whenever it is needed. Unlike rsh, there *is* an authentication transaction performed with each access; the remote host queries for the cached passphrase each time it receives a connection request.

To initialize ssh, the user must issue the ssh-keygen command on her local machine and select a passphrase. The phrase should be at least fifteen characters long and be selected with the same care as a normal password. The following session shows an example:

testuser localmachine$ **ssh-keygen**

Initializing random number generator...

Generating p: ..++ (distance 10)

Generating q: ...++ (distance 44)

Computing the keys...

Testing the keys...

Key generation complete.

Enter file in which to save the key ($HOME/.ssh/identity):

Enter passphrase (empty for no passphrase): **86 tromB0nes led*the par8**

Enter same passphrase again: **86 tromB0nes led*the par8**

Your identification has been saved in /home/testuser/.ssh/identity.

Your public key is:

1024 35
672884909892852277996781957758179897245813890591510227698436716054150446307111452163805329762550718140493130535691606574894579863699434132628821880371128867392912898947918961142413713442023184971355873040314189711227035494238385859509549949295233264167055119

69454296394218910998487385293130196466377458545747 jasmerb@koyaanisqatsi

Your public key has been saved in /home/testuser/.ssh/identity.pub

Ssh-keygen generates an identity.pub file that the user must then copy to to a file on each of the remote machines she will access. Interestingly, the file on the remote hosts must be named ~/.ssh/authorized_keys:

testuser localmachine$ **ssh -l testuser admin-101.bryant.webtv.net mkdir .ssh**

Host 'admin-101.bryant.webtv.net' added to the list of known hosts.

Password: **hRd2ges**

testuser localmachine$ **scp ~/.ssh/identity.pub testuser@admin-101.bryant.webtv.net:\ .ssh/authorized_keys**

Password: **hRd2ges**

Whenever the user expects to use ssh for authentication, she must then create a shell to handle the transactions. To do so, the *ssh-agent* and *ssh-add* commands are used:

testuser localmachine$ **ssh-agent bash**

testuser localmachine$ **ssh-add ~/.ssh/identity**

Need passphrase for /home/testuser/.ssh/identity (testuser@localmachine).

Enter passphrase: **86 tromB0nes led*the par8**

Identity added: /home/testuser/.ssh/identity (testuser@localmachine)

Subsequent ssh commands in that shell will not prompt for passwords:

testuser localmachine$ **ssh -l testuser admin-101.bryant.webtv.net date**

Tue Feb 18 17:46:23 PST 1997

InstallBuild

The WebTV installation procedures currently depend very much on the specifics of the computing environment. The scripts described here expect that certain directories will exist (e.g., $HOME/webtv, .$HOME/network, and /wtv-ops), and they assume that the login shell of user “production” on the target hosts is bash.

The key script used for software installation is *InstallBuild*. This script’s arguments include a source directory which specifies the build tree, a list of target machines, and flags which can specify ssh authentication and which service group(s) to

build. For now, InstallBuild (and the scripts described below) must be run on one of the software installation machines: admin-101 and techultra for Bryant and Alma, respectively.

The Release Control group at WebTV stores production-ready software in the ~netbuild/webtv directory on the software install machines. The source tree directory is specified using the -s option to InstallBuild. When the -s option is not specified, InstallBuild assumes that the software to be installed is in $HOME/webtv. (Similarly, it assumes that some of the tools it needs--including sdist, the ssh version of rdist--are in $HOME/network; both of these problems can be solved by running as the user "netbuild".)

The -m=*machinelist* option to InstallBuild specifies the machines to be updated. Software installers usually maintain a set of shell variables to use in the machinelist argument, so that common combinations are easily specified. The default is "all machines".

The -k option specifies that ssh should be used; the -production option specifies that the production service group's software is to be updated; the -build=*versionnumber* specifies the subdirectory in the source tree (named by a version number of the software) that contains the particular version of the software to be installed.

Finally, the -page=*address* option names the email address to which problem reports should be sent. Errors can also be captured by using the UNIX utility script to record output from the install; error messages begin with the string "ERROR".

When the installation has successfully completed, the *switch-services* command must be run on each updated machine. This command, located in the directory called ~production/webtv-*versionnumber*/tool, takes two arguments: the name of the directory which will point to the software--usually ~production/webtv--and the name of the directory which holds the current version--usually ~production/webtv-*versionnumber*. This command creates the symbolic links that point to the correct software binary and the correct directories holding the corresponding configuration, content, and tools.

The software is now installed.

The service can be initialized by running the command *start-services &* on each service server. The server will display the processes it is starting. If there are problems, error messages will usually be displayed within about twenty seconds of launch.

As a final check, the command *check-service*s should be run on the admin machine for the service. It will spawn a monitord process that ensures the right processes are running on the right machines. If its output scrolls by quickly, the service is working well. Pauses should occur only when mail is being tested. It's final message says "Done and 0 failures."

Prototype Software Installation

For the Fujitsu Trial 1.5 installation, WebTV software was loaded onto two machines named monster.japan.webtv.net and porky.japan.webtv.net. The software was to be slightly modified from the Trial 1 software in ~netbuild/

webtv-165.5; the tools, including InstallBuild, were available in ~netbuild/network-165.5. The person doing the installation was testuser; his home machine was localstation. His results are below, and comments begin with #'s.

testuser@techultra $ **ssh-agent bash**

testuser@techultra $ **ssh-add ~/.ssh/identity**

Need passphrase for /home/testuser/.ssh/identity (testuser@localstation.artemis.com).

Enter passphrase: **{passphrase entered}**

Identity added: /home/testuser/.ssh/identity (testuser@localstation.artemis.com)

testuser@techultra ; ssh -l production monster.japan.webtv.net date

Mon Feb 24 04:35:25 PST 1997

~testuser/network was a symbolic link to ~netbuild/network

testuser@techultra $ **PATH=$PATH:/home/testuser/network/src/tool; export PATH**

The following command adds sdist to the command search path.

testuser@techultra $ **PATH=$PATH:/wtv-ops/bin; export PATH**

testuser@techultra $ **~netbuild/network-165.5/src/tool/build/InstallBuild -k **
-production -m=monster.japan.webtv.net -build=165.5 -page=testuser

One command performed: ln -s /usr/home/production/webtv-165.5/config/production.solaris2.cf /usr/home/production/webtv-165.5/config/webtv.solaris2.cf

The total install time was: 11:27 (mm:ss)

testuser@techultra $ **telnet monster**

...

monster$ **cd ~production**

monster$ **ls -ld webtv***

drwxr-xr-x 5 productionother 512 Dec 11 19:26 webtv

drwxrwxrwx 6 productionother 512 Feb 24 04:57 webtv-165.5

monster$ **ls webtv**

dynamic-content spool userstore

monster$ **ls webtv-165.5**

bin.solaris2 config content tool

monster$ **~/webtv-165.5/tool/switch-services ~/webtv ~/webtv-165.5**

linking /usr/home/production/webtv/bin.solaris2 to /usr/home/production/webtv-165.5/bin.solaris2

linking /usr/home/production/webtv/config to /usr/home/production/webtv-165.5/config

linking /usr/home/production/webtv/content to /usr/home/production/webtv-165.5/content

linking /usr/home/production/webtv/tool to /usr/home/production/webtv-165.5/tool

monster$ cd /usr/home/production/webtv-165.5/config

monster$ **ls -l *.cf**

lrwxrwxrwx 1 productionother 53 Feb 24 04:57 flashrom.cf -> /usr/home/production/webtv/flashromimages/flashrom.cf

-rw-r--r-- 1 productionother 40509 Feb 5 22:21 japan1.solaris2.cf

-rw-r--r-- 1 productionother 36632 Feb 5 22:21 personal.solaris2.cf

-rw-r--r-- 1 productionother 93342 Feb 5 22:21 production.solaris2.cf

-rw-r--r-- 1 productionother 13818 Feb 5 22:21 prohibitedHumanNames.cf

-rw-r--r-- 1 productionother 61658 Feb 5 22:21 prohibitedUserNames.cf

-rw-r--r-- 1 productionother 42992 Feb 5 22:21 thrash-loadtest.solaris2.cf

lrwxrwxrwx 1 productionother 62 Feb 24 04:57 webtv.solaris2.cf -> /usr/home/production/webtv-165.5/config/production.solaris2.cf

To make the changes needed for Trial 1.5, the file japan1.solaris2.cf was copied to japan1.5.solaris.cf, and the latter was then modified. Those changes consisted of renaming CustomerDBName to the new CMDEM1, changing the hostnames to monster and porky, and changing the Warn, Fail, Warn Pager, and Fail Pager addresses to testuser.

Later, porky needed software, but unfortunately the links to ~netbuild/{network,webtv}-165.5 had been deleted. Here's how testuser handled the rest of the install:

testuser@localstation $ **cd $HOME**

testuser@localstation $ **rm network webtv**

testuser@localstation $ **ln -s ~netbuild/network ~/network**

The previous command worked because the tools in this case were not version-dependent.

testuser@localstation $ **ln -s ~netbuild/backup/165.5/webtv-165.5 ~/webtv**

testuser@localstation $ **rlogin techultra**

testuser@techultra $ **ssh-agent bash**

testuser@techultra $ **ssh-add ~/.ssh/identity**

Enter passphrase: **{passphrase entered}**

testuser@techultra $ **PATH=$PATH:/home/testuser/network/src/tool; export PATH**

testuser@techultra $ **PATH=$PATH:/wtv-ops/bin; export PATH**

testuser@techultra $ **~netbuild/network/src/tool/build/InstallBuild -k -production **

-m=porky.japan.webtv.net -build=165.5 -page=testuser

NOTE ERROR: mail.local.webtv gets installed in porky:~production/webtv-165.5, not webtv, so the chmod fails, but it looks like it got installed setuid anway...)

...

production@porky.japan.webtv.net: /home/testuser/webtv/bin.solaris2/mail.local.webtv.temp: installing

production@porky.japan.webtv.net: /home/testuser/webtv/bin.solaris2/mail.local.webtv: installing

production@porky.japan.webtv.net: special "chmod u+s /usr/home/production/webtv/bin.solaris2/mail.local.webtv"

production@porky.japan.webtv.net: chmod: WARNING: can't access /usr/home/production/webtv/bin.solaris2/mail.local.webtv

production@porky.japan.webtv.net: porky: REMOTE ERROR: shell returned 256

production@porky.japan.webtv.net: /home/testuser/webtv/bin.solaris2/mailtod: installing

...

testuser@techultra $ **rlogin porky**

porky $ **~/webtv-165.5/tool/switch-services ~/webtv ~/webtv-165.5**

InstallFlashromimages

Installing new flashrom images to be downloaded to clients is very similar to installing new software. The process is separated because new flashrom images can be provided without any interruption to the service. Currently, only three classes of machines run flashromd, so they have the only machines affected by this process. Those classes are the storefulls, storelesses, and mailtods.

Once again, this process is run from the admin machine for the service, and the files to be used are staged by the Release Control group in the ~production/flashromimages directory. There are often several versions stored in this directory.

The command *InstallFlashromimages* takes almost the same options as InstallBuild. Those options include -production, -k, and -m=*machinelist*. A new option, -cf=*configfile*, is used to determine which machines in the list will actually use the new images. InstallFlashromimages uses the sdist command to remove dated

versions of the images and to install flashrom.cf, a file used by flashromd to match service groups to image versions, and willy.tmpl, a menu of possible versions available to users in very sophisticated test service-groups.

Prototype Installation

The experience of loading flashrom images in Trial 1.5 proceeded as follows:

testuser@techultra $ **ln -s ~netbuild/flashromimages ~**

testuser@techultra $ **./InstallFlashromimages -production -k -cf=/home/testuser/japan1.5.solaris2.cf -m=monster.japan.webtv.net:porky.japan.webtv.net**

mv: cannot rename /home/testuser/flashromimages/production/version-253: Permission denied

mv: cannot rename /home/testuser/flashromimages/production/version-339: Permission denied

production@porky.japan.webtv.net: updating host porky.japan.webtv.net

production@monster.japan.webtv.net: updating host monster.japan.webtv.net

...

InstallContent

Frequently changing content is installed as the third part of an installation. Included is the content for the WebTV home page and the Explorer tree, and the filter list for age-appropriate restrictions. (Less dynamic content is included in the software release, e.g., instructions, help, and registration templates.) The Release Control group stores new content in the ~production/dynamic-content directory.

Content installation, like software installation, is a two-step process: moving the content and switching to the new version. The same command, *InstallContent*, is used for both parts, albeit with different arguments used in the two steps. The command has a form similar to InstallBuild and InstallFlashromimages:

InstallContent -build=*yyyymmdd.hhmmss* -m=*machinelist* -k -production -cf=*configfile*

This command is usually run as the netbuild user in the ~netbuild home directory; it assumes that the directory

~netbuild/dynamic-content/content-*yyyymmdd.hhmmss*/content

exists and holds the content to be installed. The InstallContent script tars and gnuzips the contents of the content directory--specified by the time it was created--into three files so that the sdist can distribute a few large files rather than take much longer to send many small ones. Currently only the storefull and storeless machine classes actually need to be updated; the config file helps InstallContent find those machines. When the files arrive, they are uncompressed and untarred in order to be installed. There is no cleanup facility, so the installer has to ensure that there is space available before the install begins. Changing this process has been labeled a high priority task.

To deploy the new content, the following command is used:

InstallContent -switchto=*yyyymmdd.hhmmss* -m=*machinelist* -k -production -cf=*configfile*

This command does an ssh to each machine, calling rename-link, which points a link in the new direction.

Custom Software

The UNIX utility sendmail is customized to support the special complications in the WebTV environment. A future revision of this document should resolve the conflicting information about whether this customized sendmail is included in the software distribution, or whether it is installed in a special process.

This document should also explain how the MX records for the domain name of the new service should be written and distributed. Currently, the webtv.net MX records point to the mailsorter machines; these lines are distributed from ns.artemis.com:

webtv.net MX 10 mailsorter-1.alma.webtv.net

webtv.net MX 10 mailsorter-2.alma.webtv.net

webtv.net MX 10 mailsorter-3.alma.webtv.net

Finally, this document should make clear how the name server code is installed and configured.

Radius Software

A Radius server runs on one of the pre-registration servers to authenticate the client when it makes contact. One authentication is performed when the box first powers up; another is done each time a user logs in. WebTV has customized the first authentication by modifying the Radius server to read digital telephony data and respond to the client with the address of an Internet Access Provider local to that client. The software is also customized to retain records on the service of the various Internet Access Providers.

The Radius software must be installed on the appropriate servers in the appropriate places; future revisions of this document should describe that process.

The software must then be configured, or modified, to respond to client connection requests on the last-resort toll-free number with IP addresses allocated to the service for those connections. The networks available for this purpose should be divided into 64-host groups, e.g., 208.74.32.0, 208.74.32.64, 208.74.32.128, and so on. Each group must then be associated with the IP address of one of the “pop-servers”--the modem racks that handle incoming calls. This mapping of pop-server IP addresses to IP address ranges for dial-in connections is called an Network Access Service (NAS) mapping. The one used in the current service is:

NAS_IP_NUMBER	ASSIGNED_IP_BASE
172.16.34.11	10.2.1.0
172.16.34.12	207.79.33.64
172.16.34.13	207.79.33.128
172.16.34.14	207.79.33.192
172.16.34.15	207.79.32.128
172.16.34.16	207.79.32.192

Note that the NAS_IP_NUMBERs do not need to be routable, nor do the IP addresses returned by the first modem rack--the one that responds to pre-registration calls.

A future revision of this section should explain how the Radius software (or is it the modem racks?) can be configured to use this mapping.

Support Software

This section may be expanded in future revisions. Possible topics include system administration support tools, security monitoring and logging, and version control for modifications made to configuration files.

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